



Ext 2

NORTH SYDNEY BOYS HIGH SCHOOL

2014

ASSESSMENT TASK 4

Mathematics Extension 2

General Instructions

- **Working time – 55 minutes**
Reading time – 5 minutes
- Write on both sides of the lined paper in the booklet provided
- Write using blue or black pen
- Board approved calculators only
- All necessary working should be shown in every question
- Each new question is to be started on a **new page**.
- Attempt all questions.

Class Teacher:

(Please tick or highlight)

- ☐ Mr Lam
- ☐ Ms Ζιάζιαρη
- ☐ Mr Ireland

Student Number:

(To be used by the exam markers only.)

Section	A	B	C	D	E	Total	Percent
Mark	$\overline{9}$	$\overline{13}$	$\overline{4}$	$\overline{9}$	$\overline{5}$	$\overline{40}$	$\overline{100}$

Section A (9 marks)

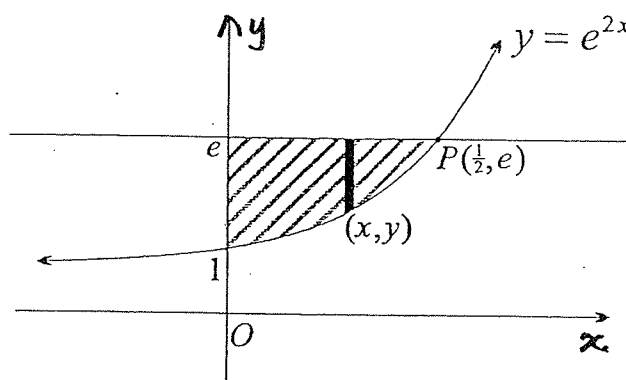
Marks

- (1) A solid has as its base the ellipse $\frac{x^2}{36} + \frac{y^2}{16} = 1$.

If each section perpendicular to the major axis is an equilateral triangle, then show that the volume of the solid is $128\sqrt{3}$ cubic units.

4

- (2) In the diagram shown below, $P(\frac{1}{2}, e)$ is the point of intersection of the curve $y = e^{2x}$ and the line $y = e$.



Using the method of cylindrical shells, find the volume of the solid generated when the shaded region enclosed by the y -axis, the curve $y = e^{2x}$ and the line $y = e$ is rotated about the y -axis.

5

Section B (13 marks)

- (3) (a) Show that $a^2 + b^2 \geq 2ab$ for any two real numbers a and b . 1

(b) Deduce that $a^2 + b^2 + c^2 \geq ab + bc + ca$ for real numbers a, b, c . 1

(c) If $a + b + c = 6$ then show that $ab + bc + ca \leq 12$ 2

(d) If a, b and c are positive, show that $(a + b)(b + c)(c + a) \geq 8abc$ 3

- (4) It can be shown that the arithmetic mean of n positive real numbers is always greater than or equal to their geometric mean. That is, if $a_1, a_2, \dots, a_n$ are

positive real numbers then $\frac{a_1 + a_2 + \dots + a_n}{n} \geq \sqrt[n]{a_1 a_2 \dots a_n}$. Using this result,

show that $n! \leq \left(\frac{n+1}{2}\right)^n$ for any positive integer $n \geq 1$. (induction is not needed) 2

- (5) (a) Given that $f(x) = \sqrt{1+x}$, show that $f'(x) < \frac{1}{6}$ for $x > 8$. 2

(b) Hence or otherwise show that $\sqrt{1+x} \leq 3 + \frac{x-8}{6}$ when $x \geq 8$. 2

Section C (4 marks)

(6) Chris believes that if $x + \frac{1}{x}$ equals an integer, say N , then $x^n + \frac{1}{x^n}$ must also be an integer, for all positive integers n .

(a) Prove that this is true for $n = 1$ and $n = 2$. 1

(b) Hence use strong induction to prove that if $x + \frac{1}{x}$ is an integer then

$x^n + \frac{1}{x^n}$ is an integer, for all integers $n \geq 1$. 3

Section D (9 marks)

(7) (a) Show that $\binom{n}{r-1} = \binom{n+1}{r} - \binom{n}{r}$, where $r = 1, 2, 3, \dots, n$ 2

(b) Hence evaluate $\sum_{n=3}^{100} \binom{n}{2}$ 2

(8) This question uses the identity $(1+x)^n = 1 + \binom{n}{1}x + \binom{n}{2}x^2 + \binom{n}{3}x^3 + \dots + \binom{n}{n}x^n$

(a) Use the sum of a geometric progression formula to simplify

$$1 + (1+x) + (1+x)^2 + (1+x)^3 + \dots + (1+x)^{n-1} \quad 1$$

(b) Hence show that

$$1 + (1+x) + (1+x)^2 + \dots + (1+x)^{n-1} = \binom{n}{1} + \binom{n}{2}x + \binom{n}{3}x^2 + \dots + \binom{n}{n}x^{n-1} \quad 1$$

(c) Find $\int_{-1}^0 \left(\binom{n}{1} + \binom{n}{2}x + \binom{n}{3}x^2 + \dots + \binom{n}{n}x^{n-1} \right) dx$ 1

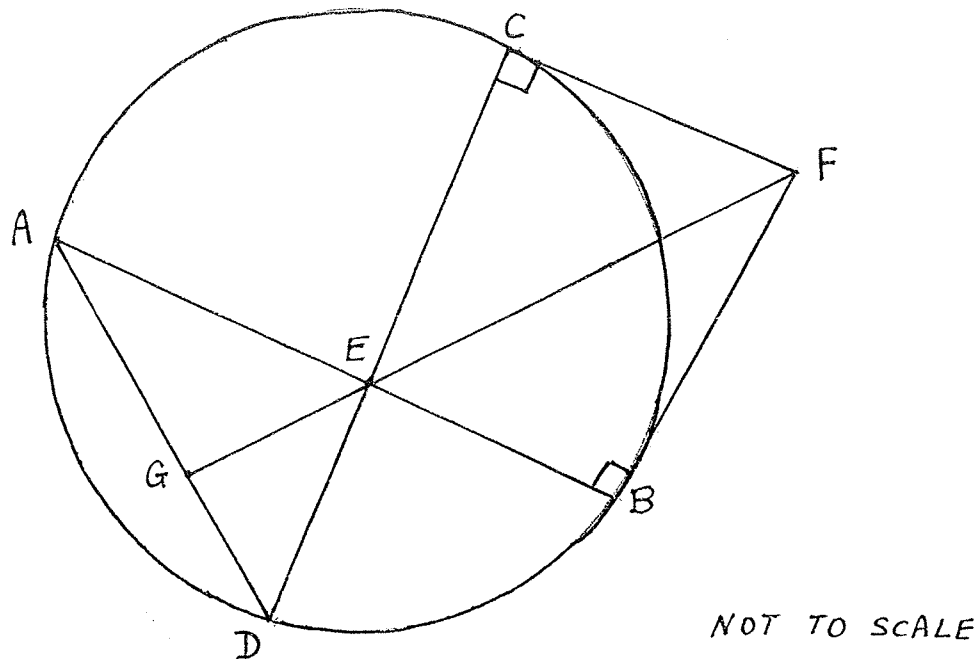
(d) Hence show that:

$$\sum_{r=1}^n \frac{(-1)^{r+1}}{r} \binom{n}{r} = \sum_{r=1}^n \frac{1}{r} \quad 2$$

Please turn over for final question →

Section E (5 marks)

(9)



In the diagram, AB and CD are two chords in a circle. AB and CD intersect at E . F is a point such that $\angle ABF$ and $\angle DCF$ are right angles. FE produced meets AD at G .

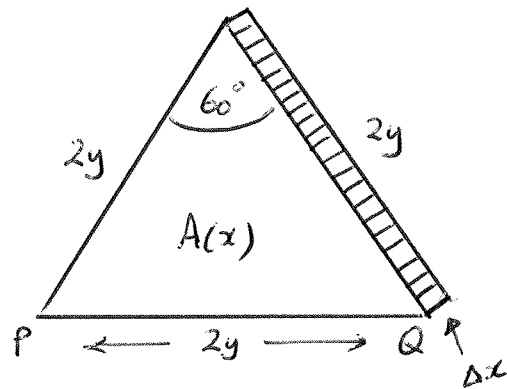
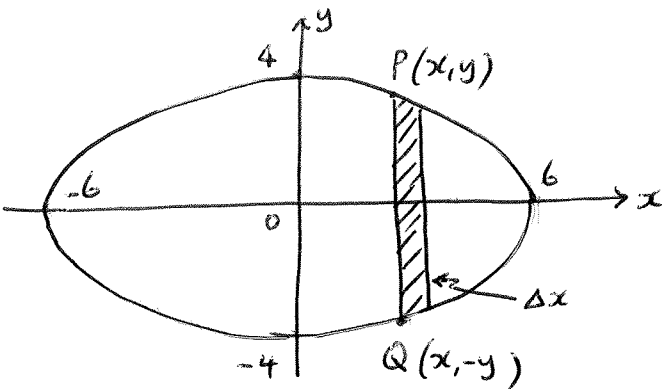
Copy the diagram into your answer booklet. (Start this question at the top of a left-hand page, so that you – and we – don't have to keep turning the page!).

- | | |
|--|---|
| (a) State why $CFBE$ is a cyclic quadrilateral. | 1 |
| (b) Hence prove that $\angle BAD = \angle GFB$ | 2 |
| (c) Hence prove that FG is perpendicular to AD . | 2 |

2014 Extension 2 - Task 4 - { SUGGESTED SOLUTIONS }

Q1

$$\frac{x^2}{36} + \frac{y^2}{16} = 1$$



$$A(x) = \frac{1}{2} \cdot 2y \cdot 2y \cdot \sin 60^\circ$$

$$\therefore A(x) = y^2 \sqrt{3}$$

$$\text{But } y^2 = 16 \left(1 - \frac{x^2}{36}\right)$$

$$\therefore A(x) = 16\sqrt{3} \left(1 - \frac{x^2}{36}\right)$$

$$\& \therefore V(x) \doteq 16\sqrt{3} \left(1 - \frac{x^2}{36}\right) \cdot \Delta x$$

$$\text{Thus } V = \lim_{\Delta x \rightarrow 0} \sum_{x=-6}^{x=6} 16\sqrt{3} \left(1 - \frac{x^2}{36}\right) \cdot \Delta x$$

$$\therefore V = 16\sqrt{3} \int_{-6}^6 \left(1 - \frac{x^2}{36}\right) dx$$

$$= 32\sqrt{3} \int_0^6 \left(1 - \frac{x^2}{36}\right) dx$$

$$= 32\sqrt{3} \left[x - \frac{x^3}{108} \right]_0^6$$

$$= 32\sqrt{3} \left[6 - \frac{216}{108} \right]$$

$$\therefore V = 128\sqrt{3} \text{ units}^3 \quad \#$$

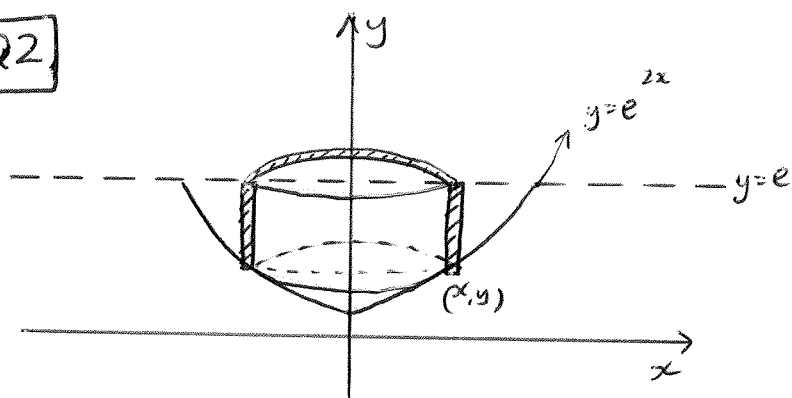
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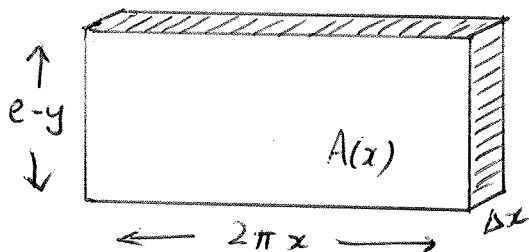
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Q2



$$\Delta V = 2\pi x(e - y) \cdot \Delta x$$



$$\therefore \Delta V = 2\pi x(e - e^{2x}) \cdot \Delta x$$

$$V = \lim_{\Delta x \rightarrow 0} \sum_{x=0}^{\frac{1}{2}} 2\pi x(e - e^{2x}) \Delta x$$

$$\therefore V = \int_0^{\frac{1}{2}} 2\pi x(e - e^{2x}) dx$$

$$\text{Now } \int_0^{\frac{1}{2}} 2\pi x e dx = 2\pi e \left[\frac{x^2}{2} \right]_0^{\frac{1}{2}} = \frac{\pi e}{4}$$

$$\begin{aligned} \text{And } \int_0^{\frac{1}{2}} 2\pi x e^{2x} dx &= 2\pi \int_0^{\frac{1}{2}} x \frac{d}{dx} \left(\frac{1}{2} e^{2x} \right) dx \\ &= 2\pi \left\{ \left[\frac{1}{2} x e^{2x} \right]_0^{\frac{1}{2}} - \int_0^{\frac{1}{2}} \frac{1}{2} e^{2x} dx \right\} \\ &= 2\pi \left[\frac{1}{2} \cdot \frac{1}{2} e' - \left[\frac{1}{4} e^{2x} \right]_0^{\frac{1}{2}} \right] \\ &= 2\pi \left[\frac{e}{4} - \left(\frac{e}{4} - \frac{1}{4} \right) \right] \\ &= \frac{\pi}{2} \end{aligned}$$

$$\text{Thus } V = \frac{\pi e}{4} - \frac{\pi}{2}$$

$$\left[\text{i.e. } V = \frac{\pi}{4}(e - 2) \right] \#$$

Q3

$$(a) \quad (a-b)^2 \geq 0$$

$$\therefore a^2 - 2ab + b^2 \geq 0$$

$$\therefore a^2 + b^2 \geq 2ab.$$

(b) Likewise,

$$a^2 + c^2 \geq 2ac$$

$$b^2 + c^2 \geq 2bc$$

Adding the 3 equations,

$$2a^2 + 2b^2 + 2c^2 \geq 2ab + 2ac + 2bc$$

$$\therefore a^2 + b^2 + c^2 \geq ab + bc + ca.$$

$$(c) \quad (a+b+c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$$

$$\therefore 6^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$$

$$\therefore 36 \geq 3(ab + bc + ca) \text{ from (b)}$$

$$\therefore ab + bc + ca \leq 12$$

(d) Since $x^2 + y^2 \geq 2xy$, x, y real (from (a))

$$\therefore a+b \geq 2\sqrt{a} \cdot \sqrt{b} \quad (\text{letting } a=x^2, b=y^2)$$

Likewise

$$b+c \geq 2\sqrt{b} \cdot \sqrt{c}$$

$$a+c \geq 2\sqrt{a} \cdot \sqrt{c}$$

$$\therefore \text{Multiplying, } (a+b)(b+c)(a+c) \geq 2\sqrt{a} \cdot \sqrt{b} \cdot 2\sqrt{b} \cdot \sqrt{c} \cdot 2\sqrt{a} \cdot \sqrt{c} \\ = 8abc$$

Q4

$$\frac{a_1 + a_2 + \dots + a_n}{n} \geq \sqrt[n]{a_1 a_2 \dots a_n}$$

Thus
$$\frac{1+2+3+\dots+n}{n} \geq \sqrt[n]{1 \cdot 2 \cdot 3 \dots n}$$

$$\text{LHS} = \frac{\frac{n(n+1)}{2}}{n} = \frac{n+1}{2}$$

$$\text{RHS} = \sqrt[n]{n!}$$

Thus
$$\sqrt[n]{n!} \leq \frac{n+1}{2}$$

$$\therefore n! \leq \left(\frac{n+1}{2}\right)^n \quad \#$$

Q5

(a) $f(x) = \sqrt{1+x} \quad \therefore f'(x) = \frac{1}{2\sqrt{1+x}}$

$$< \frac{1}{2\sqrt{1+8}} \quad \text{as } x > 8$$

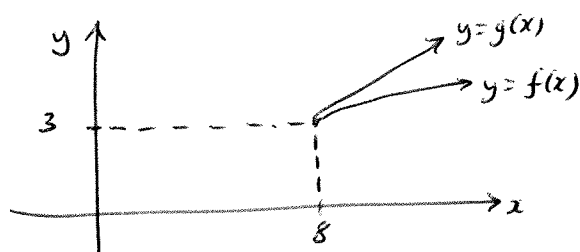
$$= \frac{1}{6}$$

(b) Let $g(x) = 3 + \frac{x-8}{6} \quad \therefore g'(x) = \frac{1}{6}$

Now $f(8) = \sqrt{1+8} = 3$
 $g(8) = 3 + 0 = 3 \quad \therefore f(8) = g(8)$

Since $f'(x) < g'(x)$ when $x > 8$,
 and $f(8) = g(8)$

$\therefore f(x) \leq g(x)$ when $x \geq 8$.



Q6

Let $x + \frac{1}{x} = N$.

(a) For $n=1$, $x^1 + \frac{1}{x^1} = x + \frac{1}{x}$
 $= N \quad \therefore \text{true for } n=1$

For $n=2$, $x^2 + \frac{1}{x^2} = \left(x + \frac{1}{x}\right)^2 - 2 \cdot x \cdot \frac{1}{x}$
 $= N^2 - 2$, an integer
 $\therefore \text{true for } n=2.$

(b) Assume $x^n + \frac{1}{x^n}$ is an integer, for all integers $1, 2, 3, \dots, k$, when $x + \frac{1}{x}$ is an integer.

Then $x^{k+1} + \frac{1}{x^{k+1}} = \left(x^k + \frac{1}{x^k}\right)\left(x + \frac{1}{x}\right) - x \cdot \frac{1}{x} - \frac{1}{x^k} \cdot x$
 $= \left(x^k + \frac{1}{x^k}\right)\left(x + \frac{1}{x}\right) - \left(x^{k-1} + \frac{1}{x^{k-1}}\right).$

But $x^k + \frac{1}{x^k}$ and $x^{k-1} + \frac{1}{x^{k-1}}$ are both integers, by our hypothesis;
 and $x + \frac{1}{x} = N$, an integer.

Thus $x^{k+1} + \frac{1}{x^{k+1}}$ is an integer.

We have proved that the statement is true for $n=k+1$ if it is true for $n=1, 2, 3, \dots, k$.

Since it's true for $n=1$ and 2 , it is true for all $n \geq 1$, by induction.

Q7 (a)
$$\binom{n+1}{r} - \binom{n}{r} = \frac{(n+1)!}{r!(n+1-r)!} - \frac{n!}{r!(n-r)!}$$

$$= \frac{n!}{r!(n+1-r)!} \cdot [(n+1) - (n+1-r)]$$

$$= \frac{n! \cdot r}{r!(n+1-r)!}$$

$$= \frac{n!}{(r-1)!(n-(r-1))!}$$

$$= \binom{n}{r-1}, \text{ as required.}$$

(b)
$$\binom{n}{2} = \binom{n+1}{3} - \binom{n}{3} \quad \text{from (a)}$$

$$\therefore \sum_{n=3}^{100} \binom{n}{2} = \sum_{n=3}^{100} \binom{n+1}{3} - \sum_{n=3}^{100} \binom{n}{3}$$

$$= \sum_{n=4}^{101} \binom{n}{3} - \sum_{n=3}^{100} \binom{n}{3}$$

$$= \binom{101}{3} - \binom{3}{3}$$

$$= 166\,650 - 1$$

$$\therefore = 166\,649. \quad \#$$

Q8

$$(a) \quad 1 + (1+x) + (1+x)^2 + \dots + (1+x)^{n-1} = \frac{1 \left[(1+x)^n - 1 \right]}{(1+x) - 1}$$

(as $r = 1+x$,
 $a = 1$
& n terms)

$$= \frac{(1+x)^n - 1}{x}$$

(b)

$$1 + (1+x) + \dots + (1+x)^{n-1} = \frac{(1+x)^n - 1}{x}$$

$$= \frac{\left[1 + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n \right] - 1}{x}$$

$$= \binom{n}{1} + \binom{n}{2}x + \dots + \binom{n}{n}x^{n-1}$$

$$(c) \quad \int_{-1}^0 \left[\binom{n}{1} + \binom{n}{2}x + \dots + \binom{n}{n}x^{n-1} \right] dx$$

$$= \left[\binom{n}{1}x + \frac{1}{2}\binom{n}{2}x^2 + \frac{1}{3}\binom{n}{3}x^3 + \dots + \frac{1}{n}\binom{n}{n}x^n \right]_{-1}^0$$

$$= \binom{n}{1} - \frac{1}{2}\binom{n}{2} + \frac{1}{3}\binom{n}{3} - \dots + \frac{(-1)^{n+1}}{n}\binom{n}{n}$$

Q8 - ctd.

(d)

$$\sum_{r=1}^n \frac{(-1)^{r+1}}{r} \binom{n}{r} = \binom{n}{1} - \frac{1}{2} \binom{n}{2} + \frac{1}{3} \binom{n}{3} - \dots + \frac{(-1)^{n+1}}{n} \binom{n}{n}$$

$$= \int_{-1}^0 \left(\binom{n}{1} + \binom{n}{2}x + \dots + \binom{n}{n}x^{n-1} \right) dx$$

$$= \int_{-1}^0 1 + (1+x) + (1+x)^2 + \dots + (1+x)^{n-1} dx$$

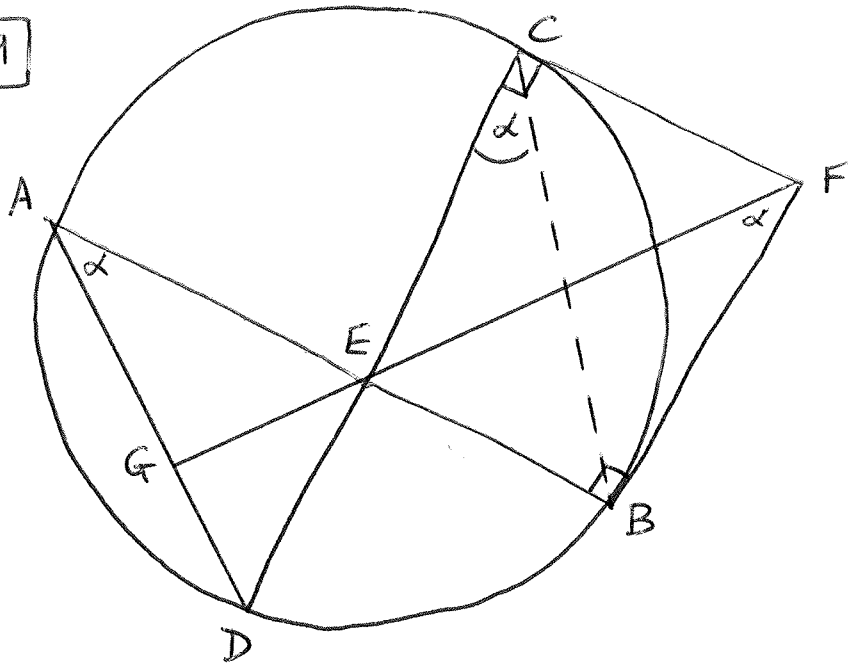
$$= \left[x + \frac{(1+x)^2}{2} + \frac{(1+x)^3}{3} + \dots + \frac{(1+x)^n}{n} \right]_{-1}^0$$

$$= \left(0 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} \right) - (-1 + 0 + 0 + \dots + 0)$$

$$= 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}$$

$$= \sum_{r=1}^n \frac{1}{r} \quad \#$$

Q9



- (a) CFBE is cyclic as opposite angles $\angle ECF$ and $\angle EBF$ are supplementary.

- (b) Draw BC ; Let $\angle ECB = \alpha$.

Then $\angle EFB = \alpha$ (angles in same segment of the circle thro $CFBE$).

i.e. $\angle GFB = \alpha$.

Also, $\angle BAD = \alpha$ (angles in same segment of original circle ACBD)

$$\therefore \angle BAD = \angle GFB.$$

- (c) Since $\angle BAD = \angle GFB$,

$\therefore$ AFBG is a cyclic quadrilateral
(BG subtends equal angles at two points,
A and F, on the same side of BG).

Hence $\angle AGF = \angle ABF$ (angles in same segment of circle AFBG)
 $= 90^\circ$. #